

TCS3351 Cryptography and Data Security

Tutorial 5 Questions

Modern Symmetric Ciphers-II (Advanced Encryption Standard (AES))

1. Answer the following questions with respect to Advanced Encryption Standard (AES).
 - a. List the block size, key size, number of rounds, number of round keys, and number of transformations for the three AES versions.
 - b. Define a state in AES. How many states are there in each version of AES?
 - c. If a and b are two bytes, show that
$$\text{SubBytes}(a \oplus b) \neq \text{SubBytes}(a) \oplus \text{SubBytes}(b)$$
Use $a = (35)_{16}$ and $b = (46)_{16}$ as an example.
2. With respect to Advance Encryption Standard, answer the following:
Given the plaintext
(00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F)_H
and the key
(01 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01)_H
 - a. Show the original contents of State, displayed as a 4×4 matrix.
 - b. Show the value of State after initial AddRoundKey.
 - c. Show the value of State after SubBytes.
 - d. Show the value of State after ShiftRows.
 - e. Show the value of State after MixColumns.

Note:

(i) SubByte Transformation Table

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	CB	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

(ii) *Constant matrices used by MixColumns and InvMixColumns*

$$\begin{array}{c}
 \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \xleftrightarrow{\text{Inverse}} \begin{bmatrix} 0E & 0B & 0D & 09 \\ 09 & 0E & 0B & 0D \\ 0D & 09 & 0E & 0B \\ 0B & 0D & 09 & 0E \end{bmatrix} \\
 C \qquad \qquad \qquad C^{-1}
 \end{array}$$

3. With reference to Advanced Encryption Standard (AES-128) key expansion algorithm, identify the first 3 subround keys.

Given Key:

(0F 15 71 C9 47 D9 E8 59 0C B7 AD D6 AF 7F 67 98)_H

References:

1. Behrouz A. Forouzan, "Cryptography and Network Security", First Edition, McGraw-Hill, 2008.
2. William Stallings, "Cryptography and Network Security – Principles and Practice", Seventh Edition, Prentice Hall, 2016.